CardioNova Health

Smarter Insights. Healthier Hearts.

AI-assisted cardiovascular risk screening | Educational demonstration project

REPORT NUMBER
CN-261105-D8482992

CARDIOVASCULAR RISK ASSESSMENT REPORT

PATIENT NAME	ASSESSMENT DATE
Sample patient (invented)	05 Nov 2026, 10:30
AGE / SEX	**REPORT STATUS**
21 years / Male	Generated automatically; not reviewed by a clinician
ASSESSMENT TYPE	**MODEL**
Machine-learning risk screening from 13 clinical inputs	Gradient Boosting (calibrated)

ESTIMATED RISK OF HEART DISEASE

LOW

9.7%

Risk meter (project-defined bands: below 30% low, 30% to below 60% moderate, 60% or more high)

Result reliability: LOW. Estimates from the model's internal versions range from 1.4% to 28.9%.

Low reliability. The category above may not be trustworthy. Please check the entered values before relying on it.

- Unusual compared with the training data: Age, Resting blood pressure, Cholesterol, Maximum heart rate.
- The cross-validation fold models disagree strongly (spread 28 percentage points).
- Maximum heart rate 62 bpm is far below the age-predicted maximum (about 199 bpm). Check that it is the peak value from the exercise test, not a resting pulse.

Summary of findings

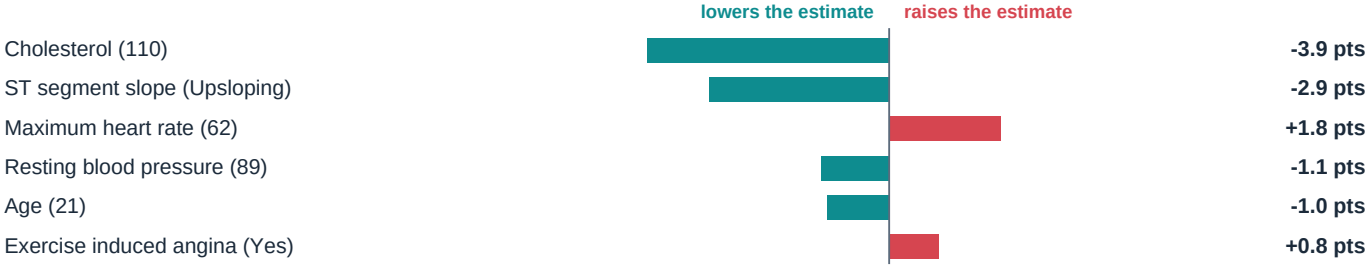
The model estimates a low probability of heart disease (9.7%) for the values entered.

- Cholesterol (110) lowered the estimate by about 3.9 percentage points compared with a typical patient in the training data.
- ST segment slope (Upsloping) lowered the estimate by about 2.9 percentage points compared with a typical patient in the training data.
- Maximum heart rate (62) raised the estimate by about 1.8 percentage points compared with a typical patient in the training data.

Guidance: This sample was built from invented values. Discuss real results with a qualified clinician.

This report presents the output of a statistical model. It does not confirm or exclude heart disease and must not be used to make treatment decisions.

Factors influencing the estimate



Each bar shows how many percentage points the estimate changes when that input is replaced by the typical value in the training data. These are model associations, not medical causes, and need not add up to the total.

Clinical parameters entered

PARAMETER	RESULT	UNIT	GENERAL GUIDE (adults)	STATUS
PATIENT				
Age	**21**	years	Risk generally rises with age	-
Sex	**Male**		Reported as entered	-
SYMPTOMS				
Chest pain type	**Non-anginal pain**		Interpreted by a clinician	-
Exercise-induced angina	**Yes**		No chest pain on exercise expected	**Present**
RESTING MEASUREMENTS				
Resting blood pressure (systolic)	**89**	mm Hg	Under 120; 120-129 borderline; 130+ high	**Below guide**
Serum cholesterol	**110**	mg/dl	Under 200; 200-239 borderline; 240+ high	**Check value**
Fasting blood sugar > 120 mg/dl	**Yes**		Fasting glucose under 100 mg/dl is usual	**Above 120**
Resting ECG	**Normal**		Normal expected	**Within guide**
EXERCISE STRESS TEST				
Maximum heart rate achieved	**62**	bpm	Age-predicted maximum about 199 bpm	**Check value**
ST depression (exercise)	**1**	mm	Under 1.0 mm is usual	**Borderline**
Slope of peak exercise ST segment	**Upsloping**		Upsloping is usual	**Upsloping**
IMAGING / LABORATORY				
Major vessels (fluoroscopy)	**0**		None (0) is usual	**None**
Thalassemia (thal) test	**Normal**		Normal is usual	**Normal**

Guide values are general adult reference points shown for context. They are not personal targets and not a diagnosis; categorical inputs have no numeric range.

Suggested next steps

- Share this report with a qualified healthcare professional and discuss it together with your symptoms, examination and tests.
- **If you have chest pain, severe breathlessness, fainting or any emergency symptom, seek emergency care immediately.** Do not wait for a report.
- Ask: What does an estimate like this mean for someone with my history and symptoms?
- Ask: Which of my test results (ECG, stress test, imaging) should be repeated or explained in more detail?
- Ask: What blood pressure, cholesterol or blood sugar targets are right for me?
- Ask: How often should I have routine check-ups?

CLINICIAN REVIEW (OPTIONAL)			
Reviewed by (name)		**Registration no.**	
Signature		**Date**	
Comments			

LIMITATIONS AND MEDICAL DISCLAIMER

About the method. Model: Gradient Boosting, trained on 297 patient records from the UCI Cleveland dataset. Held-out test ROC-AUC 0.925 (95% interval 0.851 to 0.982). The small, single-site, mostly older training population limits how well the model fits other people.

CardioNova Health is an academic machine-learning project. This report is for educational and demonstration purposes only; it is not a hospital-issued diagnostic report, does not confirm or exclude heart disease, and must not be used to make treatment decisions. The model and its probabilities have not been clinically validated for individual patient care. Consult a qualified healthcare professional for medical assessment and interpretation of clinical tests.

Project-defined risk bands: below 30% = Low; 30% to below 60% = Moderate; 60% or above = High. These thresholds are not official clinical risk classifications.